universidade de aveiro

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Roaming for Applications

M1

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125011 Margarida Pinheiro
125274 Rodrigo Sampaio
125624 Miguel Sousa
119625 João Ribeiro

2026

CONTEXT AND PROBLEM

Roaming



Edge Computing



European Initiatives

8ra CLOUD-EDGE CONTINUUM

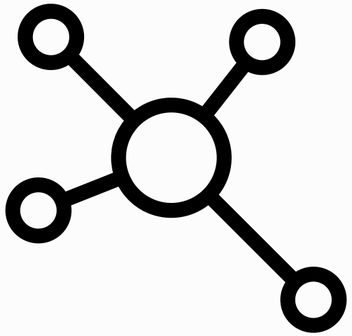
GOALS AND EXPECTED RESULTS

REQUEST



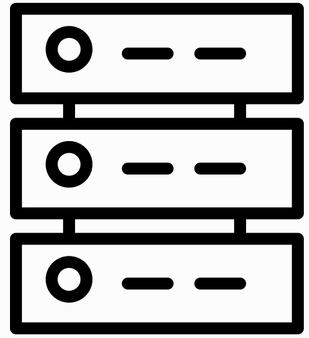
Request through one interface

DELEGATE



Automatic Partner Selection

RUN



Extend Application Coverage

Requested through Operator A. Running on Operator B. Managed from one interface.

Task List

Research (Everyone)

Task 1 (Everyone): Read IPCEI-CIS Reference Architecture (ICRA)

Task 2 (Everyone): Read GSMA v7.0 with special focus on Federation and E/WB Interfaces

Task 3 (Margarida Pinheiro): Related Work (State of the Art Draft)

Task 4 (Margarida Pinheiro): More Detailed State of the Art

Implementation Modules

Task List

Platform Infrastructure and Operator Simulation (Miguel Sousa)

Task 1 (Miguel Sousa): Request IT cloud access and resources

Task 2 (Alan Marques & Miguel Sousa): Set up Kubernetes clusters for 3 operators

Task 3 (Miguel Sousa): Write Dockerfiles for the API, FM and Deployment Layer

Task 4 (Miguel Sousa): Define simulated operators: name, locations, capacity, latency values

Federation Manager (Margarida Pinheiro)

Task 1 (Margarida Pinheiro): Define the FM's internal architecture and its interfaces to the API and Deployment Layer

Task 2 (Margarida Pinheiro & Alan Marques): Build the Deployment Layer mock so FM implementation progress can evolve independently

Task 3 (Margarida Pinheiro): Federation Catalog: data model and local publishing of the operator's offering

Task 4 (Margarida Pinheiro & Rodrigo Sampaio): Hand-over decision logic: local if capacity fits, otherwise pick a partner from its catalog

Task 5 (Margarida Pinheiro & João Ribeiro): Track application state and propagate status back to the API

Task 6 (Margarida Pinheiro): Develop tests along the way

East/West Bound Interface Implementation (Rodrigo Sampaio)

Task 1 (*Rodrigo Sampaio*): Implement the server side (endpoints exposed to partners)

Task 2 (*Rodrigo Sampaio & João Ribeiro*): Implement the client side (calls to partner FMs)

Task 3 (*Rodrigo Sampaio*): Federation onboarding messages

Task 4 (*Rodrigo Sampaio & Miguel Sousa*): Catalog exchange

Task 5 (*Rodrigo Sampaio & Miguel Sousa*): Hand-over request, status update and termination messages

Task 6 (*Rodrigo Sampaio*): Error handling: partner down, rejection, timeouts

Task 7 (*Rodrigo Sampaio*): Develop tests along the way

Implementation Modules

Task List

Developer-Facing API (João Ribeiro)

Task 1 (João Ribeiro & Alan Marques): Define the application descriptor (image, ports, resources, location, latency), shared with the Deployment Layer

Task 2 (João Ribeiro): Implement the endpoints: register, deploy, status, delete

Task 3 (João Ribeiro): Request validation and error responses

Task 4 (João Ribeiro & Miguel Sousa): Connect it to the FM

Task 5 (João Ribeiro): Generate and publish OpenAPI docs

Task 6 (João Ribeiro): Develop tests along the way

Application Deployment Layer (Alan Marques)

Task 1 (*Alan Marques & Margarida Pinheiro*): Integrate the internal interface with the FM with endpoints like deploy, status, delete, get_capacity

Task 2 (*Alan Marques*): Implement container deployment with Kubernetes (consider VM integration later)

Task 3 (*Alan Marques*): Implement lifecycle and status reporting

Task 4 (*Alan Marques*): Develop tests along the way

Project Specification Modules

Task List

Architecture (Miguel Sousa)

Task 1 (Miguel Sousa): Define the modules and their interactions

Task 2 (Miguel Sousa): Decide shared representations like location, capacity, application descriptor

East/West Bound Interface Specification (Rodrigo Sampaio)

Task 1 (*Rodrigo Sampaio*): Define the list of messages and their purpose

Task 2 (*Rodrigo Sampaio*):OpenAPI/Swagger Specification

Task 3 (*Rodrigo Sampaio*): Document the flows: onboarding, catalog exchange, hand-over, status

Developer-Facing API Specification (João Ribeiro)

Task 1 (João Ribeiro): Document endpoints and the application descriptor

Task 2 (João Ribeiro): Add request and response examples

Project Specification Modules

Task List

Technical Report (Everyone)

Task 1 (Everyone): Agree on the outline and shared template

Task 2 (Alan Marques): Draft the introduction, context and related work

Task 3 (Everyone): Everyone keeps updating the report as we do our respective tasks

Task 4 (Everyone): Final review

Project Administration Modules

Task List

Project Management and Communication (Rodrigo Sampaio & Margarida Pinheiro)

Task 1 (Margarida Pinheiro): Set up the repository and give supervisors access

Task 2 (Margarida Pinheiro): Set up the project website

Task 3 (Rodrigo Sampaio): Run regular team meetings, keep notes and manage tasks

Presentations (Everyone)

Task 1 (Margarida Pinheiro): Draft presentation template

Task 2 (Everyone): Everyone adds in the presentation their respective work for each respective milestone

Demo (Alan Marques)

Task 1 (Alan Marques): Design the demo scenario

Task 2 (Alan Marques): Dry run on the IT cloud

CALENDAR

Weeks 2–4

Week 2
22/09/2026

Team and project decided. First meeting with supervisors to understand goals and scope of project. Reading of important documentation in relation to the IPCEI-CIS and basic understanding of the current related work.

Week 3
29/09/2026

M1: Definition of context, goals, calendar, repository and tasks for each team member. First presentation done.

Week 4
06/10/2026

Better understanding of the ICRA. Investigate the ETSI SDG Open OP. Definition of requirements and use cases. Project report initiation.

CALENDAR

Weeks 5–7

Week 5
13/10/2026

M2: More precise definition of the architecture, technologies and how the Federation layer will be developed. Second presentation done.

Week 6
20/10/2026

Implementation of the Provider Access Point API and core logic for a single platform instance, with basic application deployment layer, following the architecture and technologies defined in M2.

Week 7
27/10/2026

Implementation of more platform instances, ideally 3, all with basic application deployment layers. Review/change of the calendar and objectives based on current work achieved.

CALENDAR

Weeks 8–11

Week 8 & 9
03/11/2026

MS3: Start of development of the Federation Manager and integration with the deployment layer. Draft of the EWBI API spec. Third presentation done.

Week 10
17/11/2026

Federation Manager core logic developed with discovery and onboarding between the platforms, start of the implementation of the hand-over decision logic to other partner platform instances. Tests development.

Week 11
24/11/2026

Federation Manager logic refined. Hand-over logic triggers the deployment on the partner platform. Updated tests.

CALENDAR

Weeks 12–14

Week 12 & 13
01/12/2026

Hand-over mechanisms working, including application instantiation on the partner and status reported back. Refinement and polishing of the report. Updated tests.

Week 14

15/12/2026

MS4: Review and changes in the project, current code reviewed/polished, and draft project report. Assessment of work achieved. Fourth presentation done.

STATE OF THE ART

	LIQ	GSMA	IPCEI-CIS	OpenOP by ETSI	Our Project
Inter-Operator Federation	✗	✓	✓	✓	✓
Latency & Location Aware Decision	✗	✓	✓	✗	✓
Transparent Application Handover	✗	✓	✓	✗	✓
Single Developer Entry Point (PAP)	✓	✗	✓	✓	✓
Executable Prototype	✓	✗	✗	✓	✓

Communication Plan











Our Team



Margarida Pinheiro

DevOps



Alan Marques

QA Engineer/Tester



Rodrigo Sampaio

Team Manager



João Ribeiro

Product Owner



Miguel Sousa

Architect

Advisors

Diogo Gomes & Rafael Direito

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Thank You